

vision**2**learn

Coding Practices Level 3

Hints
and
Tips

Unit 2



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Hints and Tips Unit 2

The information in this guide will be helpful to you when you are ready to complete your assessment:

Unit 2: Coding Design

Section 1 – Understand the use of coding standards

Q1. Identify the importance of coding standards:

Give a short statement for why coding standards are important.

Q2. Explain the consequences of not following coding standards. You should include how following your own untried and untested methods could adversely affect software areas such as security and performance.

Give a detailed answer that explains the consequences of not following coding standards. Think about what would happen if you didn't follow coding standards and followed your own methods, which are not agree, tried or tested. How would it affect software in areas such security and performance?

Q3. Explain the processes and tools that could be used to promote coding standards.

You will need to be able to explain processes and tools that can assist a development team in ensuring coding standards are followed. Consider manual methods and technical tooling.

Section 2 – Understand the purpose of good coding principles and practices

Q1. **Identify** good coding principles and practices used by software developers and explain why it's important that the principles and practices are followed.

Within your answer you must name different sets of coding principles that are currently used. You will need to demonstrate your understanding of the principles within each and explain why they are important. What impact do they have on the code being written and how does it benefit the developer and their peers?

Q2. In relation to KISS principle and SOLID practices, **draw** conclusions for:

- how they might change the code that you write
- the positive and negatives implications of following them
- not following the coding practices and principles

Be careful to include all the parts in this question and include both KISS and SOLID in your answer. Consider and how adhering to these guideline would change the written code. It's important that you also include both positive and negative implications of having to follow coding practices and principles, along with those of not following.

Section 3 - Know about coding reviews

Q1. Identify at least three different types of coding review that can be used in software development teams. You **must** consider the following in your answer:

- the interactions needed with each type of review
- the impact on quality and the productivity of the team
- when and why each type of review might take place

As an identify question you need to simply state your answer. Ensuring that you include information for all bullet points for each of the four different types of coding review that you identify.

Q2. Explain why code is reviewed and its importance in the development process. Include:

- what can be 'caught' during a code review
- issues that could arise if code is not reviewed
- reasons a code review might be requested or performed

You must include each bullet pointed item within your answer and ensure that you have a detailed explanation to cover why code is reviewed and the importance of doing so.

Q3. Distinguish the difference between submitting and performing a coding review, and then **explain** the importance of effective feedback that would be given in a coding review. Consider the following:

- the actions that can take place both before and after code review activities
- why it's important that the code is reviewed and not the developer
- the importance of giving effective feedback
- the important of receiving effective feedback

This is a two-part question with additional points that must be included. Your answer will therefore be detailed. It's important that a clear understanding of the difference between submitting and performing a coding review is provided, along with a detailed explanation of the importance effective feedback, and the actions that take place during and after code review activities. You should show understanding of why it's important to review the code and not the developer themselves.

- Q4. Explain the advantages and disadvantages of the following. For each one include:
- what it means
 - how it can each be integrated into the development process:
- a) Pair programming
 - b) Mob programming

This is an explain question and will therefore need a detailed answer. You need to demonstrate understanding of both programming techniques, how they can be integrated into the development process and the advantages and disadvantages brought by such activities.

Section 4 - Understand databases

- Q1. Explain what is meant by the following. Differentiate between the two types of data and how it is stored
- structured data
 - unstructured data

You must demonstrate understanding of the terms 'structured data' and 'unstructured data'. Include what types of data is classed as structured and unstructured, how each organised, stored and why it is different.

- Q2. Describe what is meant by databases. Include:
- different types of databases
 - different types of stored data
 - the difference between NoSQL and relational databases
 - today's key technologies
 - different ways of storing data

You must describe the meaning of the word 'databases', the different types of database and the data they can store. Demonstrate understanding of the difference between NoSQL and relational databases. You should also have an understanding of the key technologies in use today and know how data can be stored.

- Q3. Describe the following structure and key components of a database table:
- Field
 - Record
 - Table
 - Primary key

You should provide a detailed description for each component and could include an example of when/how it is used.

Q4. Explore the use of primary and foreign keys. Within your answer you should consider the following:

- differences between primary and foreign keys and their relationship to one another
- their qualities and purpose
- their constraints

Exploring the use of primary and foreign keys, you will need to demonstrate understanding of the difference between them, their relationship to each other, the qualities they possess and their purpose. Furthermore, you should understand the concepts of constraints and relational integrity, why primary keys must be unique and what the term 'auto-increment' means.

Q5. Describe techniques for querying databases. Include:

- query language
- different ways a query can be executed
- CRUD operations
- stored procedures

You must demonstrate understanding of query language and the concept of querying a database. You should be able to describe the different ways (and places from) a query can be executed. Plus, the basic CRUD (create, read, update and delete) operations and stored procedures.

Q6. Explain the purpose of database indexing:

This is a straightforward explain question in which you should fully justify the purpose for indexing within databases.

Section 5 - Understand the principles of interface design

Q1. Identify the elements of UX (User Experience) design and why they are important whilst designing and building user interfaces

You may provide a simple statement to identify the elements of UX design and include why they are important whilst designing and building user interfaces.

Q2. Describe the differences between UX and UI (User Interface). Include the core concepts of each

You must demonstrate understanding of the meaning of both terms (UX and UI) and the core concepts within each. Your answer should clearly demonstrate the difference between UI and UX and why they are not the same.

Q3. Explain the key skills required for creative problem-solving. Show your awareness of the role that creativity plays in creating good user experience

You will need to explain the key skills of creative problem-solving and demonstrate your awareness of the role that creativity plays in creating good user experiences.

Q4. Describe the key tools and processes that may help during the creative process and with testing new ideas

Your description must cover what key tools and processes are needed and how they may help during the creative process and with testing new ideas.

Q5. Explain the importance of information architecture (IA) and UX. Within your answer, include:

- HCI (human-computer interaction)
- why it's important
- examples of good and bad IA and reasons why

This will be a detailed answer that fully explains what is meant by HCI (human-computer interaction) and why it is important. Within your answer, you must also provide examples of good and bad IA (information architecture) and give sound reasoning as to why you consider them to be good or bad.

Q6. Explain the principles of good user interface design and how a badly designed interface can affect a product.

Within your explanation, consider the principles of good user interface design for each one and how the product may be affected by a badly designed interface.

Section 6 - Understand the importance of building security by design into software at the development stage

Q1. Describe what is meant by building security by design into software at the development stage. Include:

- how it differs to the standard SDLC (Software Development Life Cycle):
- why security should be considered as early as possible
- the potential impacts of not considering security early

You must include detail for each bullet point within your description. Ensure that you describe what security by design is and, how it differs to the standard SDLC (Software Development Life Cycle). You must also show an understanding of why security should be considered as early as possible and the potential impacts of not doing so.

Q2. Identify and explain four methods that are used to ensure software is secure.

Ensure that you name each method and then explain the key steps that is used to ensure that software is secure for each one.

Q3. Identify five types of security issues and threats that can impact software development

As an identify question you may simply name each type of security issue or threat and provide a brief statement of how each one impacts software development.